Avi Kadria

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Profile

PhD candidate in Computer Science at Bar-Ilan University, researching **distance-related graph algorithms**, with papers at **SODA, ICALP, ESA, ISAAC, and DISC**. **Completed my BSc during high school**, followed by an MSc with two SODA papers. Former IDF officer with 5+ years in Unit 8200, where I **built a new cyber R&D team from the ground up**, led critical projects, and recruited for its elite tracks; now researching full-time.

Education

Bar-Ilan University *Ramat Gan, Israel*

PhD in Computer Science *Oct 2023 - Now*

- Researching distance-related graph problems, including **distance oracles, girth, diameter, routing, and fine-grained complexity**.
- Published 7 papers in graph algorithms (listed below and at avika81.github.io/papers).
- Recipient of the **Clore Scholarship** for exceptional PhD students (2025) and the **President Scholarship** (2024).

Bar-Ilan University *Ramat Gan, Israel*

MSc in Computer Science *Oct 2017 - Mar 2023*

- Researched **girth approximations**, published and presented two papers at **SODA 2022** and **SODA 2023**.
- Collaborated with **leading global experts** in distance-related problems.
- Recipient of the **Nadav Scholarship** for exceptional research, as the sole master's student selected (2022).
- Started during the third year of the BSc, and was a full-time officer at the IDF for most of the degree.

Bar-Ilan University *Ramat Gan, Israel*

BSc in Computer Science and Pure Mathematics *Jul 2015 - Aug 2018*

- Graduated with honors at the age of 17.

Preprints

Deterministic Girth Approximation in Weighted Undirected Graphs
Avi Kadria
arXiv:2610.13452, 2026

Publications

Improved Approximation Algorithms for n-Pairs Shortest Paths
Avi Kadria, Liam Roditty, Virginia Vassilevska Williams
34th Annual European Symposium on Algorithms, ESA 2026, L'Aquila, Italy, 2026

Tighter Bounds for Weighted and Unweighted Shortest Cycle Approximation
Avi Kadria, Liam Roditty, Virginia Vassilevska Williams
34th Annual European Symposium on Algorithms, ESA 2026, L'Aquila, Italy, 2026

Faster Algorithms for (2k-1)-Stretch Distance Oracles
Avi Kadria, Liam Roditty
53rd International Colloquium on Automata, Languages, and Programming, ICALP 2026, Egham, United Kingdom, 2026

New Approximate Distance Oracles and Their Applications - **best paper award winner**
Avi Kadria, Liam Roditty
36th International Symposium on Algorithms and Computation, ISAAC 2025, Tainan, Taiwan, 2025

Compact Routing Schemes in Undirected and Directed Graphs
Avi Kadria, Liam Roditty
39th International Symposium on Distributed Computing, DISC 2025, Berlin, Germany, 2025

Improved girth approximation in weighted undirected graphs - **Accepted to SICOMP'26**
Avi Kadria, Liam Roditty, Aaron Sidford, Virginia Vassilevska Williams, Uri Zwick
Proceedings of the 2023 ACM-SIAM Symposium on Discrete Algorithms, SODA 2023, Florence, Italy, 2023

Algorithmic trade-offs for girth approximation in undirected graphs
Avi Kadria, Liam Roditty, Aaron Sidford, Virginia Vassilevska Williams, Uri Zwick
Proceedings of the 2022 ACM-SIAM Symposium on Discrete Algorithms, SODA 2022, Virtual Conference / Alexandria, VA, USA, 2022

Awards

2026 **Rector's Award for Outstanding Students**, Computer Science *BIU*
2026 **2nd place, Feder Family Award for Best Student Work in Communications**, Computer Science *Israel*
2025 **Best Paper Award, ISAAC 2025**, Computer Science *Taiwan*
2025 **1st place, Israeli Bughouse Championship (2018, 2024, 2025)**, Chess *Israel*
2022 **3rd place, Bughouse World Championship**, Chess *Germany*
2018 **IMO-trial and IOI-trial Olympiad Excellence**, Mathematics and Computer Science *Israel*
2011 **Ranked 5th in Israel (Under 14)**, Chess *Israel*

Military Service

Unit 8200 *Tel Aviv, Israel*
Cyber R&D Team Lead *May 2022 - May 2024*
- Formed a new team of Cyber R&D developers to solve crucial problems in the core of the unit's operations.
- Multidisciplinary team operating at cyber security and intelligence interface. Worked in **Python, C, ASM, HTML/JS, and SQL**.
- Mentored new team members, built the team's training program, and led numerous cross-team projects.

Unit 8200 *Tel Aviv, Israel*
Recruitment officer for ARAM, GA'ASH, and ARAZIM *Aug 2021 - May 2022*
- Oversaw and redesigned the entire recruitment process for the most elite tracks of Unit 8200, conducting over 500 interviews.

Unit 8200 *Tel Aviv, Israel*
R&D engineer *Feb 2019 - Aug 2021*
- Practical cyber security operations in Windows environments and large networks. Worked mainly in **Python and C++**.
- Developed new systems and optimized existing methods to increase efficiency and achieve outcomes previously unattainable.

Work Experience

D.S.E *U.S.A*
Algorithms Engineer *May 2018 - Feb 2019*
- Applied linear programming to develop and optimize employee scheduling algorithms, collaborating with multi-country teams.

Skills and Interests

Programming Python, C, C++, Rust, ASM, (passable in Java, C#, HTML, JS, SQL, MATLAB, etc.)
Miscellaneous Hebrew (native), English (fluent), French (passable). Interests: chess, bughouse, reading.